

SIMRAN KAUR

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EDUCATION

Boston University Boston, MA
Bachelor of Science, Biomedical Engineering (Concentration in Machine Learning) May 2026

- **Awards and Honors:** Spring 2023 Undergraduate Research Program Awardee, BUild Lab First-Year Innovation Fellowship
- **Biomedical Engineering Senior Design Project (Senior Capstone)**
 - Title: Modeling miRNA-siRNA crosstalk via regulatory networks to optimize siRNA therapeutics
 - Project Advisor: Hao Chen, Ph.D. (Amgen)

Relevant Coursework: Introduction to Machine Learning | Probability, Statistics & Data Science | Principles of Molecular Cell Biology and Biotechnology | Systems Physiology | Systems Biology & AI | Biomedical Signals & Controls | Modern Control in Biological Systems | Computational Biology and Machine Learning | RNA Therapeutics and Personalized Medicine (Spring 2026) | Clinical Applications of Biomedical Design (Spring 2026)

SKILLS

Programming: R (igraph, tidyverse, e1071, RGCCA), MATLAB, Python (numpy, matplotlib, scipy.signal), C, Bash, Git

High-performance computing: Slurm and PBS systems, genome sequence alignment, TargetScan, PLINK

Statistical Analysis: High dimensional mediation analysis, ANOVA, PCA, regression and correlation analysis, gene set enrichment analysis

Data: SNP genotype array, RNAseq (bulk), miRNA expression, ICD9/10 codes, phecodes

Cohort Studies/EHR Systems: ROSMAP, Inovalon medical claims

Machine Learning: Clustering (e.g. k-means), local outlier factor, isolation forests, one-class support vector machine

Biological Databases: KEGG, GO, LINCS, Connectivity Map (Clue.io), FlyBase

Data Visualization: Heatmaps, violin plots, tree maps, scatter plots, dendrograms, network diagrams

Laboratory: Immunofluorescence, immunohistochemistry, confocal microscopy, fly husbandry and genotyping, *Drosophila* ovarian dissection

Editing: Microsoft Office, Canva, Adobe InDesign

EXPERIENCE

Amgen Boston, MA
Boston University Senior Project Team Member Sept 2025 - Present

- Collaborated as part of computational team to understand miRNA-siRNA crosstalk to optimize siRNA therapeutics
- Created regulatory graph network to aid in understanding miRNA-siRNA crosstalk to guide future siRNA design and mitigate off-target effects

Harvard Medical School Department of Biomedical Informatics Boston, MA
Summer Institute Intern Jun 2025 - Aug 2025

- Utilized unsupervised machine learning methods (anomaly detection) to identify and diagnose undiagnosed pediatric patients from over 1 million Inovalon medical claims data
- Created pipeline in R to filter patients and create frequency matrix of patients and ICD9/10 codes; applied dimensionality reduction by converting ICD codes to phecodes
- Discovered over 9 clusters of anomalous patients; created visualization media to understand cluster-driven phenotypes of anomalous patients
- Presented findings to faculty and researchers from HMS DBMI in 15-minute presentation

The Jackson Laboratory Bar Harbor, ME
Summer Research Fellow Jun 2023 - Aug 2023

- Created pipeline on identifying transcriptomic signatures to mediate effect of genotype on phenotype in Alzheimer's disease using multivariate causal inferencing method on cognitively aging population (ROSMAP)
- Analyzed multi-omics data (genotype, transcriptome, phenotype) in R with high-dimensional mediation analysis to obtain scores of up and down-regulated genes of causal importance in Alzheimer's disease

- Performed downstream analyses on ranked gene list such as gene enrichment analysis, gene ontology analysis, and drug repositioning to understand biological pathways and perturbagens relevant in Alzheimer's disease
- Compiled research and functional enrichment analysis to understand biological themes represented in Alzheimer's disease and their causal influences from gene interactions
- Presented findings to principal investigators and researchers at The Jackson Laboratory's 2023 Summer Student Program Symposium

Neurovascular Imaging Laboratory at Boston University

Boston, MA

Research Assistant

Aug 2024 - Oct 2024

- Constructed mouse visual cortex machine learning simulation using Allen Institute V1 model
- Fine-tuned foundational modeling of cortical column through debugging algorithm and generated raster plots of neurons through command line prompts
- Created various visual media on MATLAB Psychtoolbox to increase robustness and generalization of simulation

McCall Lab at Boston University Department of Biology

Boston, MA

Undergraduate Researcher

Oct 2022 - Jul 2024

- Studied cell clearance and death in *Drosophila melanogaster* oogenesis; organized all lab fruit fly lines and monitored sick strains weekly
- Operated confocal microscopy, used immunohistochemistry and genotyping to analyze staining patterns
- Investigated role of microtubules on V-ATPases membrane-trafficking in stretch follicle cells

PUBLICATIONS

- **Kaur S**, Tyler AL, Durante GL, Cary GA, Carter GW and Mahoney JM (2026) Identifying transcriptomic signatures that mediate the causal effect of genotype on Alzheimer's disease. *Front. Neurosci.* 20:1716828. doi: 10.3389/fnins.2026.1716828

RESEARCH FUNDING

National Science Foundation Award No. 2244034

- Awarded to support research at Research Experiences for Undergraduates (REU) with The Jackson Laboratory Summer Student Program under Dr. Matthew Mahoney

Dr. Susanne E. Churchill Summer Institute in Biomedical Informatics (SIBMI) Summer Stipend

- Awarded with fully funded summer research stipend to support research at Harvard Medical School Department of Biomedical Informatics SIBMI Program under Dr. Shilpa Nadimpalli Kobren and Dr. Zak Kohane

ADDITIONAL PROJECTS

Bayesian Network Modeling of Cell-Cell Communication for Drug Repositioning in Triple-Negative Breast Cancer

- Collaborated in group to construct a probabilistic knowledge graph that integrates ligand-receptor interactions, pathway activities, and drug-target relationships to model intercellular signaling dynamics in TNBC
- Quantitatively analyzed inferences for the probability that a drug causes a particular effect on a cell type

LEADERSHIP

Boston University Engineering Student Government

Sept 2022 - Present

Vice-President

- Served as liaison between student government and students to facilitate social and professional events per year; collaborated with administration and constituents of College of Engineering to foster collaborative environment
- Led College of Engineering in discourse with Dean of Engineering to address areas for improvement

The Beacon

Oct 2023 - Present

Vice-President and Associate Editor

- Promoted student voices through oldest social literary journal on campus as inclusive space for creative works
- Hosted 3+ open mic nights, guest poet readings, writing workshops every semester
- Analyzed over 100+ submitted works for acceptance into released journal issue per semester; led staff editors and graphic design teams through review process